FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Multiple Pregnancy

Lecture | Week 4 · Hypertension, twins & the pelvic floor | Obstetrics | Dr Ndiwalana

Twins are common in Uganda and high-risk for mother and babies. Chorionicity (not zygosity) determines the risks, so establish it by ultrasound in the first trimester (lambda vs T sign), then give enhanced ANC, watch for preterm labour, pre-eclampsia, anaemia, TTTS and growth discordance, and plan delivery in a hospital with theatre and an experienced team ready to manage the second twin and PPH.

Must know before the ward round

1. Chorionicity decides risk. Dichorionic = lambda (twin-peak) sign; monochorionic diamniotic = T sign. Best seen at 11–14 weeks; if uncertain, manage as MC.
2. All dizygotic twins are DCDA. Monozygotic twins split at 0–3 days = DCDA, 4–8 days = MCDA (commonest MZ type), 8–12 days = MCMA, >13 days = conjoined.
3. Scan surveillance - DC every 4 weeks from 20 weeks; MC every 2 weeks from 16 weeks (DVP, bladders, EFW, Doppler).
4. Discordance = (larger EFW − smaller EFW) ÷ larger EFW × 100; ≥25% (some use ≥20%) is significant - divide by the LARGER twin.
5. TTTS affects 10–15% of MCDA twins - unbalanced placental vascular anastomoses; donor DVP <2 cm (stuck twin), recipient DVP >8 cm and heart failure. Quintero staging; laser ablation at 16–26 weeks.
6. Twins need extra iron/folate (Hb at booking, 20–24 and 28 weeks) and aspirin 75–150 mg from 12 weeks if one more pre-eclampsia risk factor (many give it to all twins).
7. Timing (NICE) - uncomplicated DCDA at 37+0 weeks, MCDA at 36+0 weeks, MCMA by CS at 32+0–33+6 weeks, triplets at 35+0 weeks.
8. Vaginal birth is reasonable if twin 1 is cephalic; CS if twin 1 is non-cephalic, MCMA, or triplets.
9. Twin labour needs theatre, 2 units cross-matched, IV access, both FHRs monitored and two neonatal resuscitation sets.
10. After twin 1 - NO oxytocic; check the lie of twin 2, stabilise it longitudinally, check the FHR, then ARM when the presenting part enters the pelvis. Oxytocin infusion if contractions are weak.
11. Give the third-stage uterotonic only after the LAST baby; prepare for PPH (overdistended uterus) with a prophylactic oxytocin infusion.
12. Locked twins - twin 1 breech, twin 2 cephalic, chins interlock. Prevent by CS when twin 1 is breech.

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1. Definition

A **multiple pregnancy** is one with **two or more fetuses**. Twins are by far the commonest; triplets and higher-order multiples are rare and usually follow fertility treatment.

Term	Meaning
Zygosity	The number of **fertilised eggs** (genetics): **dizygotic (DZ)** = two eggs + two sperm ("non-identical", "fraternal"); **monozygotic (MZ)** = one egg splits ("identical")
Chorionicity	The number of **placentas (chorions)**: **dichorionic (DC)** or **monochorionic (MC)**
Amnionicity	The number of **amniotic sacs**: **diamniotic (DA)** or **monoamniotic (MA)**
Discordant growth	EFW difference ≥20–25% of the larger twin's weight: (larger EFW − smaller EFW) ÷ larger EFW × 100
Selective FGR (sFGR)	One twin's EFW <10th centile, usually with discordance ≥25%
TTTS	Twin-twin transfusion syndrome (section 4)
TAPS	Twin anaemia-polycythaemia sequence: tiny anastomoses → one twin anaemic, the other polycythaemic, **without** liquor discordance
TRAP	Twin reversed arterial perfusion: an **acardiac twin** perfused by the "pump" twin
Vanishing twin	Early loss of one twin, usually in the first trimester
Locked twins	Twin 1 breech, twin 2 cephalic; the chins interlock above the pelvic brim

Chorionicity, not zygosity, determines risk. Determine it by ultrasound early (ideally 11–14 weeks). The types (DCDA, MCDA, MCMA, conjoined) are in section 5.

The one thing never to forget: **after the first twin is born, do not give oxytocin/ergometrine for the third stage** (you would trap twin 2). **Check the lie of the second twin** and be ready to act.

2. Epidemiology

- **Monozygotic twinning is constant worldwide** at about **3.5–4 per 1,000 births**: it is a random event, not inherited, and not affected by race or age.
- **Dizygotic twinning varies widely**, driven by FSH levels (multiple ovulation). It is highest in **sub-Saharan Africa** (often around 18–28 per 1,000 births; the Yoruba of Nigeria and parts of Benin are the famous examples), intermediate in Europe/USA, and lowest in East Asia. **Sub-Saharan Africa has the highest twinning rates in the world**, so you will see twins on every Ugandan labour ward.
- Because DZ twins predominate in Africa, the **proportion of MC twins is lower** in Uganda than in Europe (where about 1 in 3 twins are MC). But MC twins still occur and carry most of the fetal risk.
- Twins account for a disproportionate share of **perinatal deaths** (perinatal mortality several times higher than for singletons) and **preterm births** (about half of twins deliver before 37 weeks).
- Twins are **high-risk for everyone**: the mother has more **anaemia, pre-eclampsia, preterm labour, malpresentation, operative delivery and PPH**; the babies have more **prematurity, growth restriction, anomalies and intrapartum hypoxia** (especially **twin 2**), and **monochorionic twins** share a placenta, which brings unique complications (TTTS, TAPS, co-twin death).

UGANDA CONTEXT

Cultural context: among the Baganda, twins (*abalongo*) are special. There are traditional names (e.g. **Babirye/Nakato** for girls, **Waswa/Kato** for boys) and ceremonies, and the mother is called *Nalongo* and the father *Salongo*. Families are usually delighted, but twins bring real **financial strain** (feeding, transport for ANC, hospital delivery costs). Uganda MoH policy is that **twins should be delivered at a facility with capacity for caesarean section and newborn care (HC IV or hospital)**, not at HC II/III or at home. Undiagnosed twins (no scan) still present in labour at lower-level facilities, and a "retained second twin" is a classic referral.

3. Aetiology and Risk Factors

Dizygotic twins come from multiple ovulation (raised FSH). **Monozygotic twins** come from a random split of one embryo.

Risk factors for DZ twins

Non-modifiable	Treatment-related
Black African ethnicity	**Ovulation induction** (clomiphene, gonadotropins)
Maternal family history (the mother's side matters, because it is about her ovulation)	**IVF** (multiple embryo transfer)
Older maternal age (peaks at 35–39 years, as FSH rises)	
High parity	
Tall/heavier women	
Previous DZ twins (DZ twinning tends to recur)	

MZ twinning has no known risk factors: constant worldwide, not inherited, not affected by race or age.

4. Pathophysiology

Why twins cause more maternal problems

Physiological change (exaggerated in twins)	Consequence
Higher hCG	More **hyperemesis gravidarum**
Greater plasma volume expansion (about 500 mL more than singletons) + **higher iron and folate demand**	**Anaemia** (dilutional and iron/folate deficiency)
Larger placental mass → more sFlt-1	**Pre-eclampsia** 2–3 times more common, earlier and more severe
Larger placental mass → more human placental lactogen	**Gestational diabetes**
Uterine overdistension → gap junctions, oxytocin receptors	**Preterm labour, PPROM**
Large placental surface area	**Placenta praevia**
Sudden decompression after twin 1	**Placental abruption** before twin 2
Overdistended uterus contracts poorly + large placental bed	**PPH** (atonic)
Increased cardiac output and pressure on organs	Breathlessness, back pain, oedema, varicose veins, reflux

Physiological change (exaggerated in twins)	Consequence
Two fetuses competing for space	**Malpresentation** (especially twin 2), cord prolapse, locked twins

Why monochorionic twins are special: the shared placenta

In **every MC placenta** the two fetal circulations are connected by **vascular anastomoses** on the chorionic plate:

- **Artery-to-artery (AA)** and **vein-to-vein (VV)** anastomoses: superficial and **bidirectional**. They balance blood flow (AA anastomoses are protective against TTTS).
- **Artery-to-vein (AV)** anastomoses: deep, through a shared cotyledon, and **unidirectional**: blood from twin A's artery drains into twin B's vein.

TTTS develops when **unbalanced AV anastomoses** send net blood from one twin (the **donor**) to the other (the **recipient**) without enough AA compensation:

Donor twin	Recipient twin
Hypovolaemia → renin-angiotensin activation	**Hypervolaemia** → atrial natriuretic peptide release
Oliguria → **oligohydramnios** (deepest pocket <2 cm): the "**stuck twin**" plastered to the uterine wall	**Polyuria** → **polyhydramnios** (deepest pocket >8 cm)
Small or absent bladder	Large bladder
Growth restriction, anaemia	**Cardiac overload** → cardiomyopathy, tricuspid regurgitation, **hydrops**
Abnormal umbilical artery Doppler	Abnormal ductus venosus Doppler

The polyhydramnios also causes **preterm labour and PPROM**. Without treatment, severe mid-trimester TTTS has a perinatal mortality of **80–90%**.

Why co-twin death is so dangerous in MC twins: when one MC twin dies, its circulation becomes a low-pressure sink. The surviving twin **exsanguinates acutely into the dead twin** through the anastomoses → hypotension, brain ischaemia. The survivor has about a **15% risk of death** and about a **25% risk of neurological injury**. In DC twins (no anastomoses) the risks are much lower (about 3% and 2%), and the main risk is preterm labour.

MCMA twins share one sac: **cord entanglement** is almost universal, so there is a risk of sudden unpredictable death. Hence early planned **caesarean** at 32–34 weeks.

Locked twins happen when twin 1 is **breech** and twin 2 is **cephalic**: as twin 1's body delivers, its chin locks against twin 2's chin above the pelvic brim.

Multiple ovulation (DZ) or embryo split (MZ)	big	tiny
↓		
Two or more fetuses, larger placental mass		
↓		
C	Uterine overdistension	Monochorionic: shared placenta with anastomoses
↓		
Hyperemesis, GDM, pre-eclampsia, anaemia	Preterm labour, PPROM, malpresentation, atonic PPH	H
Co-twin death: survivor exsanguinates, brain injury		
↓		
Polyhydramnios, hydrops, FGR, fetal death		

Feature links

- Two fetuses + larger placenta → uterus large for dates (fundal height ≥3–4 cm above dates), globular abdomen.
- Two fetuses → multiple poles, many fetal parts, a head small for the uterine size, two fetal hearts differing by ≥10 bpm.
- High hCG → hyperemesis in early pregnancy.
- Plasma volume expansion + iron/folate demand → anaemia → pallor, tiredness, dizziness, breathlessness.
- Large placental mass → sFlt-1 → pre-eclampsia → hypertension, proteinuria, headache, epigastric pain (earlier and more severe).
- Overdistension → contractions, leaking liquor, pelvic pressure before 37 weeks (preterm labour, PPROM).
- Recipient twin polyuria → polyhydramnios → rapidly enlarging tense abdomen, breathlessness (TTTS).
- Donor twin oliguria → oligohydramnios, stuck twin, small or absent bladder, FGR.
- Space competition → malpresentation of twin 2, cord prolapse, locked twins.
- Decompression after twin 1 → abruption → bleeding and twin 2 bradycardia.
- Overdistended uterus + large placental bed → atony → PPH after the last baby.

5. Classification / Staging

By chorionicity and amnionicity

Type	Placentas and sacs	Who	Specific risks
DCDA	Two placentas (may be fused), two sacs	**All DZ twins** and about 25–30% of MZ twins	The lowest risk
MCDA	One placenta, two sacs	About 70–75% of MZ twins	**TTTS, TAPS, sFGR**
MCMA	One placenta, one sac	About 1–2% of MZ twins	**Cord entanglement** and sudden death
Conjoined twins	Incomplete division of the embryonic disc (after day 13)	Rare	Delivery by CS; specialist care

Timing of the split in monozygotic twins

Day of splitting after fertilisation	Stage	Resulting type	% of MZ twins
Day 0–3	Morula, before chorion forms	DCDA	~25–30%
Day 4–8	Blastocyst, chorion formed	MCDA	~70–75%
Day 8–12	After amnion forms	MCMA	~1–2%
Day 13+	Embryonic disc	Conjoined twins	Rare

Memory aid: "the later the split, the more they share". First they share nothing, then the chorion, then the amnion, then the body.

> EXAM FAVOURITE
> Examiners love this: **"Are all DCDA twins non-identical?"** No. All DZ twins are DCDA, but about a quarter of MZ twins are also DCDA (early split). **Different-sex twins are always DZ.** Same-sex DCDA twins may be either, and only DNA testing proves zygosity. **All MC twins are MZ.**

TTTS: Quintero staging

Stage	Findings
I	Oligohydramnios in donor (deepest pocket <2 cm) + polyhydramnios in recipient (>8 cm); **donor bladder still visible**
II	**Donor bladder not visible**
III	**Critically abnormal Doppler** in either twin (absent/reversed end-diastolic flow in the umbilical artery, reversed a-wave in the ductus venosus, pulsatile umbilical vein)
IV	**Hydrops** in either twin
V	**Death** of one or both twins

MC sFGR is classified by umbilical artery Doppler (Gratacós types I–III).

Presentation at delivery: cephalic/cephalic (about 40%), cephalic/non-cephalic (about 35–40%), non-cephalic twin 1 (about 20%) (section 9).

6. Clinical Features

History

Twins present in two ways: at **ANC** (large for dates, or a scan report saying "twins") and on the **labour ward** (in labour, bleeding, hypertensive, or after one baby has already been born somewhere else). The workup always answers: **How many babies? What chorionicity? Is the mother safe (BP, Hb, bleeding)? Are both babies well and growing equally? Where, when and how will they be born?**

Ask	Why it matters (what answer changes the plan)
Dating: LNMP, earliest scan (use the **larger twin's CRL** for dating)	Delivery dates depend on GA (DCDA 37, MCDA 36 weeks); steroids if preterm
Has any scan reported chorionicity? At what gestation?	The whole care pathway depends on it; a report after 20 weeks is less reliable. **Unknown = manage as MC**
Family history of twins (mother's side), **fertility treatment** (clomiphene, IVF)	DZ twinning is inherited through the mother; triplets after treatment
Hyperemesis in the first trimester	Supports multiple pregnancy (higher hCG)

Ask	Why it matters (what answer changes the plan)
Rapidly increasing abdominal size, breathlessness, tightness (16–26 weeks)	**Polyhydramnios** → in MC twins, **TTTS** (urgent specialist scan)
Headache, visual symptoms, epigastric pain, swelling	**Pre-eclampsia** (earlier and more severe)
Tiredness, dizziness, breathlessness; iron/folate adherence	**Anaemia** (greater demand); pulmonary oedema
Contractions, leaking, pelvic pressure before 37 weeks	**Preterm labour/PPROM** → steroids, $MgSO_4$ <32 weeks, transfer
Bleeding (painless or painful)	Praevia (large placental area) or abruption
Fetal movements	Hard to judge with two, but a change matters (fetal death, sFGR)
Previous CS, previous PPH, parity	Mode of delivery; PPH preparedness
HIV status, syphilis, Rh group, Hb at booking	eMTCT plan; anti-D; anaemia
Aspirin, calcium, IPTp taken?	Pre-eclampsia and malaria prevention
Distance from hospital, money, support	Admission near term (e.g. 36 weeks); birth preparedness

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), with the obstetric abdominal examination in its standard sequence.

Introduction, consent and preparation

- Introduce yourself; confirm her name, age and gestation; explain and gain consent; **offer a chaperone**.
- **Hand hygiene**.
- **Position and exposure**: semi-recumbent with a left lateral tilt (a large twin uterus causes marked aortocaval compression); expose from the xiphisternum to the symphysis.

General inspection and vital signs

- **Pallor** (anaemia), breathlessness, **oedema**, weight.
- **BP** at every visit (pre-eclampsia), pulse, temperature (PPROM, chorioamnionitis), RR; **shock index** (HR ÷ SBP; **≥0.9** abnormal) if bleeding.
- Signs of heart failure or breathlessness (polyhydramnios, anaemia).

Hands, face and neck

- **Hands**: palmar pallor; oedema of the fingers.
- **Face**: conjunctival pallor; facial oedema (pre-eclampsia).
- **Neck**: brief; JVP if breathless.

Obstetric abdominal examination

Step	Finding	What it means
Inspection	Abdomen larger than dates, often **globular**; tense, shiny skin	Twins, polyhydramnios
Fundal height	**Larger than dates by ≥3–4 cm**	Differential: wrong dates, polyhydramnios, macrosomia, fibroids, molar pregnancy (early)
Lie (palpation)	Count the **poles** (two heads, or three or more poles), **multiple small parts**, a head that seems **small for the size of the uterus**; state the **lie of each twin**	Twins

Step	Finding	What it means
Presentation	Presentation of each twin; which one is lower (**twin 1 = the presenting twin**)	Twin 1 non-cephalic → CS
Engagement (fifths palpable)	Of twin 1	Progress
Liquor and tone	Tense uterus with a **fluid thrill** = polyhydramnios; **uterine tenderness** (abruption, chorioamnionitis); contractions	TTTS, abruption, preterm labour
Auscultation	**Two fetal hearts** heard at the same time by two examiners at different points, differing by **≥10 bpm**	Supports twins but **not reliable alone**: confirm by ultrasound

Vaginal examination

In labour, after praevia is excluded: dilatation, membranes, presenting part of **twin 1**, station, cord.

In the second stage after twin 1 is born: palpate again: **lie and presentation of twin 2**, stabilise it longitudinally, **FHR of twin 2**, then a **VE** (membranes, presenting part, cord). After **every** delivery, palpate the abdomen before giving oxytocin: a uterus that is still large is another fetus until proven otherwise.

Completing the examination

- Bedside ultrasound (section 8); urine dipstick.
- **Thank** her, cover her, **wash hands**; **summarise**.

Document the key negatives: "BP 118/72, no pallor, SFH 38 cm at 32 weeks, twin 1 cephalic 3/5, twin 2 breech, two FHRs 140 and 128, uterus soft and non-tender, no bleeding".

Red flags

RED FLAG

- **Rapidly enlarging abdomen, breathlessness, tense uterus** in an MC twin pregnancy → **TTTS** (urgent scan)
- **Hypertension, proteinuria, headache, epigastric pain** → pre-eclampsia (earlier and more severe in twins)
- **Contractions, leaking liquor, pelvic pressure** before 37 weeks → preterm labour/PPROM
- **Hb <10 g/dL** (and worse, <7) → anaemia needing treatment
- **Reduced fetal movements**, **discordant growth ≥25%**, abnormal Doppler, single fetal death
- In labour: **transverse lie of twin 2**, **cord prolapse**, **fetal bradycardia of twin 2**, **bleeding after twin 1** (abruption), **undiagnosed second twin** after a delivery at a lower-level facility
- After delivery: **heavy bleeding** (atonic PPH)

RED FLAG

Before giving the third-stage uterotonic after any birth, **palpate the abdomen**. A uterus that is still large after the baby is out is a second twin until proven otherwise.

Thinking like a clinician

- **"Large for dates" is twins until proven otherwise** in Uganda, if there has been no scan. Get an ultrasound.
- **Ask one question about any twin pregnancy: "What is the chorionicity?"** It determines scan frequency, the delivery date, and what can go wrong.

- **Anticipate the complications in order**: early (hyperemesis, miscarriage, vanishing twin, anomalies), middle (TTTS, anaemia, preterm labour), late (pre-eclampsia, growth discordance, malpresentation), labour (twin 2 problems), postpartum (PPH, breastfeeding difficulty, depression).
- **Twin 2 is the vulnerable one in labour**: after twin 1 is born, the uterus shrinks and the placenta may partly separate, the cord can prolapse and the lie can change. Twin 2 has more hypoxia, operative delivery and death, especially if the **inter-twin interval** is long.
- **Differential for "polyhydramnios in twins"**: TTTS (MC only, with an oligohydramnios co-twin), diabetes, anomalies (oesophageal atresia), infections. A **big difference in liquor between MC twins is TTTS until proven otherwise.**
- **Discordant growth in DC twins** is usually placental insufficiency of one placenta; in MC twins it is sFGR (unequal placental sharing) or TTTS.
- **Make sure you are not counting the same heart twice**: identify each twin's heart and check the maternal pulse too.
- **Common student mistakes**: forgetting to palpate for a second baby before giving oxytocin; giving ergometrine or oxytocin after twin 1; calling "same sex = identical"; forgetting that pre-eclampsia risk means **aspirin** may be indicated; not preparing **two neonatal resuscitation sets**; forgetting iron and folate.

7. Differential Diagnoses

Of "uterus large for dates"

Differential	Distinguishing clinical feature	Distinguishing investigation
Multiple pregnancy	Many fetal parts, ≥2 poles, two FHRs	**Ultrasound**
Wrong dates	Uncertain LNMP, size consistent with a later GA	Dating/biometry scan
Polyhydramnios	Tense, fluid thrill, fetal parts hard to feel	Ultrasound: AFI >24–25 cm or deepest pocket >8 cm
Fetal macrosomia	Diabetes, large single fetus	Ultrasound EFW, glucose
Fibroids complicating pregnancy	Irregular firm masses, pain (red degeneration)	Ultrasound
Molar pregnancy (first/second trimester)	Bleeding, hyperemesis, early pre-eclampsia, no fetal parts	Ultrasound, β-hCG
Full bladder, ovarian mass	Mass separate from uterus	Ultrasound

Of complications specific to MC twins

Differential	Distinguishing clinical feature	Distinguishing investigation
TTTS	Oligohydramnios (deepest pocket <2 cm) in donor + polyhydramnios (>8 cm) in recipient	Ultrasound (deepest pockets, bladders, Doppler)
sFGR	One twin small (<10th centile), liquor normal or mildly low in both	EFW, umbilical artery Doppler
TAPS	Large difference in Hb without liquor discordance	**Middle cerebral artery peak systolic velocity** (MCA-PSV) Doppler
TRAP	One normal "pump" twin + an acardiac mass with retrograde flow	Ultrasound, colour Doppler

8. Clinical Workup

Bedside

Test	How / what to look for	What it changes	Available at
Point-of-care ultrasound	Number of fetuses, **presentation of twin 1**, **lie of twin 2**, viability, placental site, gross liquor difference	Mode of delivery (**twin 1 non-cephalic → CS**); praevia → no VE; big liquor difference in MC twins → TTTS	HC IV and above
FHR of both twins (Pinard/Doppler; dual-channel CTG where available)	Two different rates, each identified (label twin 1 and twin 2)	Fetal well-being; make sure you are not counting the same heart twice (check the maternal pulse too)	All levels (CTG at hospitals)
Hb (HemoCue)	**<11 g/dL** anaemia; **<7** severe	Iron, transfusion planning, PPH reserve	HC III and above
Urine dipstick	Protein (pre-eclampsia), nitrites/leucocytes (UTI), glucose	PE work-up; treat UTI (preterm labour trigger)	All levels
BP at every visit	≥140/90	Pre-eclampsia pathway	All levels
RBS	Raised	GDM screen (commoner in twins)	HC III and above (variable)
mRDT	Positive	Treat malaria (preterm labour, anaemia)	All levels
Bedside clotting test	No clot by 7 min	Abruption, retained dead twin, pre-eclampsia with bleeding	All levels
Partograph	As for singletons, from active labour	Labour progress for twin 1; the inter-twin interval is recorded separately	All levels

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
Full blood count / Hb	**Booking, 20–24 weeks and 28 weeks** (more often if low), and before delivery	Hb <11 (twin demand for iron and folate)	Treat with 120 mg elemental iron daily; plan blood for delivery	All HC IVs and hospitals
Blood group, Rh, antibody screen	Booking; **group and cross-match 2 units** for delivery	Rh negative	Anti-D; blood ready for PPH	HC IV with blood bank upward
HIV, syphilis, HBsAg	Booking and retest per protocol	Positive	eMTCT; infant prophylaxis for **both** babies	All levels (rapid tests)
Pre-eclampsia bloods (FBC, LFT, creatinine)	If hypertensive or proteinuric	Platelets <100, AST >70, creatinine >90	Severe PE pathway; delivery	Hospitals

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
First-trimester scan (number, viability, **chorionicity/amnionicity**, CRL dating, NT)	**11–14 weeks** (ideally; any first scan)	**Lambda sign** = DC; **T sign** = MCDA; no membrane = MCMA; different sexes = DZ (DC)	Sets the surveillance schedule and delivery date. **Uncertain = manage as MC**	HC IV (sometimes), regional/national referral, private
Anomaly scan	**18–22 weeks**	Cardiac and neural tube anomalies (commoner in MZ twins)	Counselling; place of birth	Regional/national referral, private
Growth scans: EFW each twin, **deepest vertical pocket (DVP)**, bladders, umbilical artery Doppler	**DC: every 4 weeks from 20 weeks. MC: every 2 weeks from 16 weeks**	**Discordance ≥25%** (≥20% is used by some) or one twin **<10th centile** = significant; DVP **<2 cm** donor / **>8 cm** recipient = TTTS	Doppler surveillance, fetal medicine referral, steroids, earlier delivery	Regional/national referral; basic EFW at many HC IVs
Ultrasound on admission in labour	Every twin labour	Presentation of both; lie of twin 2	Mode of delivery; manoeuvres for twin 2	HC IV and above

Ultrasound signs of chorionicity

Sign	Appearance	Meaning
Lambda (λ) / twin-peak sign	Triangular tongue of placental (chorionic) tissue extending into the base of the inter-twin membrane	**Dichorionic**
T sign	Thin membrane meeting the placenta at a right angle with no tissue between	**Monochorionic diamniotic**
No dividing membrane	Single sac; cords may be entangled	**Monoamniotic**
Two separate placentas	Two distinct placental masses	Dichorionic
Different sexes		Dizygotic → dichorionic
Membrane thickness	Thick (>2 mm, 4 layers: amnion-chorion-chorion-amnion) vs thin (<2 mm, 2 layers: amnion-amnion)	DC vs MC (less reliable later)

> **CLINICAL PEARL**
> Chorionicity is **most accurate at 11–14 weeks** (the lambda sign fades as the pregnancy advances, and after 20 weeks it can be impossible to be sure). If chorionicity is uncertain, **manage as monochorionic** (the higher-risk assumption).

Labelling: label the twins (e.g. **twin 1 = the lower/presenting twin**, or "left/right" with a description) consistently on every scan, so that growth trends are tracked for the right baby.

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
MCA peak systolic velocity (MCA-PSV) Doppler	MC twins from about 20 weeks; any single fetal death in MC twins	Classic criteria: **>1.5 MoM** in one twin (anaemia) and **<1.0 MoM** in the other (polycythaemia) = **TAPS** (newer criteria use the difference between the twins)	Fetal medicine referral; delivery planning	Specialist centres only
OGTT (75 g)	24–28 weeks if risk factors/available	Fasting ≥5.1, 1 h ≥10.0, 2 h ≥8.5 mmol/L (WHO 2013)	GDM management	Hospitals
Urine culture	Booking or symptoms	Asymptomatic bacteriuria	Treat (preterm labour trigger)	Referral labs
Clotting profile / fibrinogen	Single fetal death (baseline and periodically), abruption	Falling fibrinogen, prolonged PT/APTT (rare)	DIC risk; plan delivery	Regional/national referral
Placental examination (± histology)	After every twin birth	Number of placentas; **thin translucent membrane** (MC) vs thick opaque (DC)	Confirms chorionicity; audit; counselling for future	Examination everywhere; histology at referral hospitals
MRI of the survivor's brain	3–4 weeks after an MC co-twin death	Ischaemic injury	Counselling	Where available

UGANDA CONTEXT

Most Ugandan women with twins are first diagnosed on a **third-trimester scan** or in labour, when chorionicity can no longer be read reliably. Ask for an **early scan (before 14 weeks)** when you suspect twins at booking; if a woman arrives later with uncertain chorionicity, **treat as MC** (2-weekly scans at a referral hospital, delivery at 36 weeks) unless the twins are clearly of **different sexes** or there are **two separate placentas**.

Investigation timeline (what should have been done by the time you meet her)

Gestation	DCDA twins	MC twins (or chorionicity uncertain)
11–14 weeks	Dating (larger CRL), **chorionicity**, NT; booking bloods (Hb, group, HIV, syphilis, HBsAg), urine	Same; **label the twins**
16 weeks		Start **2-weekly scans**: DVP each sac, bladders (TTTS)
18–22 weeks	Anomaly scan	Anomaly scan (include the fetal heart)
20 weeks onwards	**Growth scan every 4 weeks** (EFW, DVP, discordance)	**Every 2 weeks**: EFW, DVP, bladders, umbilical artery Doppler; **MCA-PSV** where available (TAPS)
20–24 and 28 weeks	**Hb** (more often if low); OGTT if indicated	Same
Every visit	BP, urine protein, symptoms of preterm labour	Same, plus rapid abdominal growth or breathlessness (TTTS)
Before delivery	Presentation scan, Hb, **group and cross-match**	Same

Putting it together

Model one-line summary

> "A 30-year-old G4P3 at 32+2 weeks with a **DCDA twin pregnancy** (lambda sign at 12 weeks), twin 1 cephalic, twin 2 breech, **growth discordance of 12%**, normal liquor and Dopplers, Hb 9.8 g/dL, BP normal, for iron optimisation, 4-weekly growth scans and planned vaginal birth at 37+0 weeks."

Problem list

1. Multiple pregnancy: number, **chorionicity/amnionicity** (and how reliably known), GA.
2. Fetal: growth of each twin, discordance, liquor (DVP each sac), Dopplers, presentations.
3. Maternal: anaemia, BP/pre-eclampsia, GDM, preterm labour/PPROM risk.
4. Delivery plan: date (by chorionicity), place (theatre, blood, two neonatal teams), mode (presentation of twin 1), plan for twin 2.
5. Postpartum: PPH prevention, breastfeeding two babies, KMC, family planning.

Working diagnosis: always state number + chorionicity + amnionicity + GA + complication, e.g. "MCDA twins at 22 weeks with Quintero stage II TTTS" or "DCDA twins at 35 weeks, twin 1 cephalic, in early labour".

Model case presentation (ward round)

> "Mrs N.R., a 33-year-old para 4+0 (all alive), at 35+1 weeks by an 11-week scan, with a **DCDA twin pregnancy** confirmed by the lambda sign at 12 weeks, admitted at 34 weeks for birth preparedness because she lives 50 km away. There is a family history of twins on her mother's side. She is on iron/folate twice daily and aspirin 150 mg until 36 weeks. Latest Hb 10.4 g/dL, BP 124/78, urine protein negative. Scan at 34 weeks: twin 1 cephalic, EFW 2.3 kg; twin 2 transverse, EFW 2.1 kg (discordance 9%), normal liquor and Dopplers. **Plan**: induction of labour at 37+0 weeks in the main labour ward with an experienced obstetrician, two neonatal teams, cross-matched blood and an oxytocin infusion ready; plan ECV or internal podalic version for twin 2 if still transverse after twin 1 delivers; prophylactic oxytocin infusion after delivery; recheck Hb."

Worked example 1: reading a chorionicity scan

Scan finding at 12 weeks	Interpretation
Two gestational sacs, thick membrane with a **triangular tongue of placenta at its base**	**Lambda sign → DCDA**
One placental mass, **thin membrane meeting it at a right angle**	**T sign → MCDA**
One placental mass, **no dividing membrane**, cords close together	**MCMA** (confirm; plan CS at 32–33+6 weeks)
Scan at 26 weeks: "single placenta, thin membrane, same sex"	**Cannot be sure** (two placentas can fuse and the lambda sign fades): **manage as MC**

Worked example 2: calculating growth discordance. EFW twin 1 = **2,400 g**; EFW twin 2 = **1,800 g**.

Discordance = (larger − smaller) ÷ larger × 100 = (2,400 − 1,800) ÷ 2,400 × 100 = 600 ÷ 2,400 × 100 = 25%.

Reading it: 25% is **significant** (the threshold used in these notes is ≥25%, with some units acting at ≥20%). Next: is the smaller twin **<10th centile** (sFGR)? What is the **umbilical artery Doppler** of the small twin? What is the **chorionicity**? In DC twins, increase surveillance (Doppler every 1–2 weeks) and deliver when the small twin's Doppler deteriorates or at the planned date, balancing prematurity for the healthy twin; give steroids if delivery before 34 weeks is likely. In MC twins, refer to fetal medicine (sFGR or TTTS). **Remember to divide by the LARGER twin's weight**, not the smaller one (600 ÷ 1,800 = 33% is a common exam error).

Worked example 3: MC twins with unequal liquor. MCDA twins at 20 weeks: twin A DVP **1.5 cm**, bladder visible; twin B DVP **9 cm**, large bladder; Dopplers normal; no hydrops.

Reading it: DVP **<2 cm** in one sac and **>8 cm** in the other in MC twins = **TTTS, Quintero stage I** (donor bladder still visible). (Some fetal medicine centres use **>10 cm after 20 weeks** for the recipient; follow the referral centre.) **Action:** urgent referral to a fetal medicine specialist, weekly (or more frequent) scans to detect progression to stage II (donor bladder not visible), counselling on laser ablation (16–26 weeks; not routinely available in Uganda) or amnioreduction for symptomatic polyhydramnios, and steroids when viability approaches.

9. Management

Immediate / emergency

First 5 minutes (labour ward)

1. **A/B/C – vital signs and shock index** (HR ÷ SBP; **≥0.9** abnormal): twins mean more **APH, PPH and anaemia**. BP for **pre-eclampsia**; temperature (PPROM, chorioamnionitis).
2. **Is she bleeding?** Antepartum: think **praevia** (no VE until scan) or **abruption**. Postpartum: **atonic PPH** → call for help, massage, oxytocin (only once all babies are out), TXA, the E-MOTIVE bundle.
3. **Has a baby already been born?** A woman arriving after delivering one baby at home or at an HC III with a large uterus = **retained second twin**: **no oxytocic**, check lie and FHR of twin 2, prepare to deliver (red flag box below).
4. **Is she in labour, and how far?** Contractions, bearing down, visible presenting part. Late preterm or term twins in advanced labour are delivered **where you are**, with help called: two resuscitation sets, oxytocin drawn up but **not** given.
5. **Both fetal hearts**: two examiners or a Doppler/ultrasound, identifying which heart belongs to which twin. **Twin 2 bradycardia after twin 1** = expedite.
6. **Quick bedside scan** (if available) for **number, presentation of twin 1, lie of twin 2, placental site**: this decides vaginal birth vs caesarean.
7. **Call**: the most experienced obstetrician available, midwife, anaesthetist (theatre may be needed for twin 2), and the neonatal team. **IV access (16–18 G)**, Hb and **group and cross-match 2 units**.
8. At HC II/III with undiagnosed twins in early labour: **refer** (twins should be born where there is theatre). If delivery is imminent, deliver twin 1, **no oxytocic**, and refer at once if twin 2 is transverse or delayed.

> RED FLAG
> **"Retained second twin"**: a woman who delivered one baby at home or a HC III and still has a large uterus with a second fetus. **Do not give oxytocics before delivery of twin 2.** Assess the lie and FHR; if longitudinal and the cervix is open, deliver vaginally (ARM, oxytocin infusion if needed). If transverse, if the cervix has clamped down, if there is fetal distress or sepsis, or if **ruptured uterus** is suspected → caesarean section. If twin 2 is dead and transverse with a retraction ring, CS is still usually safest in modern practice.

Definitive

Antenatal care (enhanced ANC)

1. **Confirm chorionicity** on a first-trimester scan (11–14 weeks). If unavailable or uncertain, **treat as MC** and refer for a specialist scan.
2. **Refer to a hospital-based ANC clinic** (obstetrician-led; HC IV at a minimum) and plan delivery there.
3. **More frequent visits**: at least every 2–4 weeks from 20 weeks, and every 1–2 weeks from 32 weeks (more for MC twins).

4. **Nutrition and supplements** (drug table): iron + folic acid, calcium, low-dose aspirin, IPTp-SP, Td vaccine, ITN, deworming.
5. **Screen** at each visit: BP, urine protein, symptoms of preterm labour, Hb at 20–24 and 28 weeks.
6. **Ultrasound surveillance** (DC 4-weekly from 20 weeks; MC 2-weekly from 16 weeks).
7. **Counsel** about danger signs (preterm labour, bleeding, pre-eclampsia symptoms, reduced movements) and **birth preparedness** (money, transport, blood donor, where to deliver). Consider **admission near term** (e.g. 36 weeks) for women who live far from the hospital.
8. **Preterm birth prevention**: routine cervical cerclage and routine progesterone **do not** help in twins; bed rest does not help. Treat UTIs and malaria. Give **dexamethasone** if preterm birth is likely between 24 (28) and 34 weeks.

TTTS

- **Fetoscopic laser ablation** of the placental anastomoses (Solomon technique) at **16–26 weeks** is the treatment of choice for stage II and above (and some stage I). It improves survival (at least one survivor in about 85–90%) and reduces neurological damage compared with amnioreduction.
- **Serial amnioreduction** (draining the recipient's polyhydramnios) relieves maternal symptoms and reduces preterm labour; it is an option where laser is unavailable.
- **Selective reduction** (cord occlusion) in selected severe cases.
- Stage I may be monitored closely (it can remain stable or regress).
- After 26 weeks: steroids and delivery if worsening.

> **UGANDA CONTEXT**
> Fetoscopic laser surgery is **not routinely available in Uganda**. In practice, suspected TTTS at a Ugandan hospital means: urgent review by an obstetrician with good ultrasound skills, counselling, **amnioreduction** for symptomatic polyhydramnios where expertise exists, **steroids** when viability approaches, close surveillance and timely delivery. Some families seek referral abroad. This reality is why **early chorionicity scanning and 2-weekly MC surveillance** matter: they allow early counselling.

Discordant growth / sFGR

- **DC twins** with discordance ≥25% or one twin <10th centile: Doppler surveillance of the small twin; deliver when the small twin's Doppler deteriorates, balancing the risk of prematurity for the healthy twin.
- **MC sFGR**: classified by umbilical artery Doppler (Gratacós types I–III); referral to fetal medicine; the risk is sudden death of the small twin, with harm to the co-twin through the anastomoses.

Single fetal death

	DC twins	MC twins
Risk to the survivor	Mainly **preterm labour**	**Death (~15%)** and **neurological damage (~25%)** from acute exsanguination into the dead twin
Management	Expectant; aim for term or at least 37 weeks; watch for preterm labour	Referral to fetal medicine; **immediate delivery does NOT prevent the damage** (it happens at the moment of death); MRI of the survivor's brain 3–4 weeks later where available; plan delivery at 34–36 weeks
Maternal DIC	Rare (theoretical with retained dead fetus >4 weeks); check clotting	Same

Timing of delivery

Type	Planned birth (uncomplicated, NICE NG137)	Mode
DCDA	**37+0 weeks**	Vaginal if twin 1 is cephalic
MCDA	**36+0 weeks** (after steroids per senior decision)	Vaginal if twin 1 is cephalic
MCMA	**32+0 to 33+6 weeks** (after steroids)	**Caesarean section** (cord entanglement)
Triplets	**35+0 weeks**	Usually **caesarean section**

Continuing beyond these dates increases the risk of **stillbirth**. Earlier delivery may be needed for pre-eclampsia, FGR, TTTS, PPROM, etc.

Mode of delivery

Presentation	Recommendation
Cephalic / cephalic (about 40%)	**Vaginal delivery**
Cephalic / non-cephalic (about 35–40%)	**Vaginal delivery** is reasonable with an experienced obstetrician (twin 2 can be turned or delivered by breech extraction). The **Twin Birth Study** (2013) showed planned CS gave **no benefit** over planned vaginal birth when twin 1 is cephalic (32–38 weeks)
Non-cephalic twin 1 (breech or transverse, about 20%)	**Caesarean section** (risk of **locked twins** if breech/cephalic)
MCMA, conjoined twins, triplets	**Caesarean section**
Other indications	Same as singletons (placenta praevia, previous classical CS, two or more previous CS, severe fetal compromise); **one previous CS** is not an absolute indication but most Ugandan units favour repeat CS

Intrapartum management of twins: step by step

Before delivery (preparation):

1. **Confirm the presentation of both twins** by ultrasound on admission.
2. Deliver in a **facility with theatre** and blood (HC IV/hospital), in a room close to theatre.
3. **Senior obstetrician** (or the most experienced person available), a **midwife**, the **anaesthetist informed**, and **two neonatal resuscitation teams/sets** (two resuscitaires, two sets of towels, cord clamps labelled "Twin 1" and "Twin 2").
4. **Wide-bore IV access** (16–18 G), FBC, **group and cross-match 2 units**.
5. **Monitor both fetal hearts** (continuous CTG with two channels where available, otherwise intermittent auscultation of both, identifying each).
6. **Epidural analgesia** (where available) is helpful: it allows manipulations for twin 2 and rapid CS.
7. Have an **oxytocin infusion prepared** (but not running) for augmentation after twin 1, and ready for PPH prevention after twin 2.
8. Partograph as usual; **oxytocin augmentation** in the first stage can be used with caution if contractions are inadequate.

Delivery of twin 1:

9. Deliver **twin 1** as for a singleton cephalic delivery.
10. **Clamp and cut the cord**; label **"Twin 1"** (one clamp on the maternal end; some units tie one ligature). Hand the baby to the neonatal team.
11. **DO NOT give oxytocin, ergometrine or misoprostol** at this point.

Delivery of twin 2 (the critical phase):

12. **Palpate the abdomen** (and/or use ultrasound) to determine the **lie and presentation of twin 2**. **Listen to the fetal heart** of twin 2.
13. Have an assistant **stabilise the lie** from the abdomen (hands on either side of the uterus, holding twin 2 in a longitudinal lie).
14. **Do a vaginal examination**: check membranes (intact?), presenting part, and exclude **cord prolapse**.

Lie of twin 2	Action
Longitudinal, cephalic	Wait for the head to **descend into the pelvis** (with contractions). When the head is engaged/in the pelvis, **ARM with a contraction** (not before, to avoid cord prolapse). Encourage pushing. If contractions are weak after about **10 minutes**, start **oxytocin infusion** (e.g. 2.5–5 IU in 500 mL, titrated). Vacuum if the head is low and there is delay or fetal distress
Longitudinal, breech	Allow descent; ARM when the breech is in the pelvis; **assisted breech delivery**. If fetal distress: **breech extraction**
Transverse / oblique	1) **External cephalic version (ECV)** to cephalic (or to breech) with membranes intact, if the FHR is normal. 2) If ECV fails or is not possible, an experienced obstetrician performs **internal podalic version** (grasp a foot through intact membranes, bring it down) followed by **breech extraction**. 3) Otherwise **caesarean section**

15. **Fetal distress in twin 2** or **cord prolapse**: expedite (vacuum/forceps if cephalic and low; breech extraction; otherwise **emergency caesarean**).
16. **Inter-twin interval**: there is no absolute limit if twin 2's FHR is monitored and normal, but fetal acidosis increases with time; historically **aim to deliver within 30 minutes**. At a facility without continuous monitoring, don't let it drift.
17. Deliver twin 2; clamp the cord with **two clamps** (labelled **"Twin 2"**).

Third stage:

18. After the **last** baby: **oxytocin 10 IU IM/IV** (active management of the third stage), controlled cord traction.
19. **Prophylactic oxytocin infusion** (e.g. 20–40 IU in 1 L normal saline over 4 hours) is common practice because of the high PPH risk.
20. **Examine the placenta(s)** and membranes: count, completeness, and **chorionicity** (a thin, translucent dividing membrane suggests MC; confirm by histology where available). Label and record.
21. **Monitor closely for PPH** for at least the next 1–2 hours (fundus, lochia, pulse, BP).

Locked twins

- **What**: twin 1 **breech**, twin 2 **cephalic**; as twin 1's trunk delivers, its chin locks under twin 2's chin (chin-to-chin) above the pelvic inlet.
- **How rare**: roughly **1 in 1,000 twin deliveries** (much rarer overall), but with a very high perinatal mortality for twin 1.
- **Prevention**: **caesarean section when twin 1 is not cephalic** (the main reason for this rule).
- **Management if it happens**: call for senior help; under general anaesthesia (deep relaxation), an assistant pushes twin 2's head upward abdominally while the operator tries to disengage the chins and deliver twin 1 (rotating the body and flexing twin 1's head). If this fails and twin 1 is **alive**: **emergency caesarean section** (twin 2's head is dislodged from above; twin 1 may be delivered through the abdomen). If twin 1 is **dead**: historically, **decapitation** of twin 1 to deliver twin 2, but in modern practice **caesarean section** is usually preferred.

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Ferrous sulphate (200 mg = 60 mg elemental) + **folic acid 400 µg**	1 tablet (many clinicians give **twice daily** in twins or if Hb <11); treatment of anaemia **120 mg elemental iron daily**	Oral	Daily	Throughout pregnancy and 3 months postpartum	Greater iron and folate requirement in twins
Calcium	1.5–2 g elemental	Oral	Daily	Throughout pregnancy	Pre-eclampsia prevention in low-calcium diets
Aspirin (low-dose)	**75–150 mg**	Oral	At night	**From 12 weeks until 36 weeks** (NICE NG133 continues it until birth; many LMIC protocols stop at 36 weeks)	Multiple pregnancy is a **moderate risk factor**; give if there is **one more** moderate risk factor (e.g. first pregnancy, age ≥40, BMI ≥35). Many clinicians give it to all twin pregnancies in high-burden settings
IPTp-SP	Standard dose	Oral	Monthly from 13 weeks	Until delivery	Not with cotrimoxazole
Td vaccine, ITN, deworming (mebendazole)	Routine				Mebendazole after the first trimester
Dexamethasone	**6 mg**	IM	12-hourly	**4 doses**	If birth likely at 24 (28)–34 weeks; before planned MCDA (36 weeks, senior decision) and MCMA birth
Oxytocin (augmentation for twin 2)	2.5–5 IU in 500 mL, titrated (check unit protocol)	IV infusion	Continuous	Until twin 2 delivered	Prepared but not running until after twin 1; start if contractions weak after about 10 min
Oxytocin (third stage)	**10 IU**	IM/IV	Once	**Only after the LAST baby**	Never after twin 1
Oxytocin (PPH prophylaxis)	**20–40 IU in 1 L** normal saline	IV infusion	Over 4 hours	After delivery	High PPH risk
Tranexamic acid	1 g	IV	Once (repeat per PPH protocol)	If bleeding	
Ergometrine / misoprostol	Per PPH protocol				**Never before the last baby is delivered**
Anti-D	Per local protocol	IM			Rh-negative women
Thromboprophylaxis (LMWH)	Per local protocol	SC	Daily	After CS or with other risk factors	

Referral criteria

Level	Actions	Refer up when
HC II/III	Suspect twins (large for dates, two FHRs); health education; iron/folate, IPTp. In labour with undiagnosed twins: deliver twin 1 if imminent, **no oxytocic until twin 2 is delivered**; if twin 2 is longitudinal and descending, deliver	**All twins**: refer for ultrasound and hospital ANC. In labour: if twin 2 is transverse or delayed, **refer urgently** with IV access and an escort
HC IV / General Hospital	Enhanced ANC, basic ultrasound, planned delivery with theatre and blood; manage twin 2 (ECV, breech delivery, CS); PPH management; KMC	MC twins with complications (TTTS, sFGR), MCMA, triplets and very preterm twins → referral hospital
Regional Referral Hospital	Specialist ultrasound (chorionicity, Doppler), planned CS for MCMA/triplets, NICU	MCA-PSV, amnioreduction, fetal medicine expertise not available locally → NRH
National Referral Hospital	Specialist ultrasound (MCA-PSV), amnioreduction, NICU, fetal medicine referral (laser abroad)	Laser ablation is not routinely available in Uganda

Split between RRH and NRH adapted from the source notes, which grouped them together Verify *which RRHs offer Doppler and amnioreduction*.

Monitoring and follow-up

- **Antenatal**: BP and urine every visit; Hb at booking, 20–24 and 28 weeks; scans DC 4-weekly from 20 weeks, MC 2-weekly from 16 weeks; DC discordance → Doppler every 1–2 weeks; TTTS stage I → weekly (or more frequent) scans.
- **Intrapartum**: both FHRs; partograph for twin 1; inter-twin interval recorded.
- **Postpartum**: fundus and lochia every 15 minutes for the first hour, then half-hourly, for at least 1–2 hours.

Postpartum care

1. **PPH vigilance**: fundus and lochia every 15 minutes for the first hour, then half-hourly; oxytocin infusion; TXA 1 g IV if bleeding.
2. **Check Hb** before discharge (anaemia is common); continue iron and folate for 3 months.
3. **Breastfeeding support**: both twins can be breastfed (simultaneously in the "rugby/football hold"); milk supply responds to demand. Early and exclusive breastfeeding.
4. **Newborn care**: low birth weight/preterm: **KMC** (one twin on each side of the chest is possible), thermal care, feeding support, and early review of any growth-restricted or anaemic/polycythaemic twin (TAPS).
5. **Screen for postpartum depression** and fatigue; mobilise family support.
6. **Family planning**: counsel before discharge; offer LARC; interpregnancy interval ≥24 months. DZ twinning tends to recur (the woman has a higher chance of twins again).
7. **Thromboprophylaxis** after CS or with other risk factors.

10. Complications

Disease-related

Maternal	Fetal / neonatal
Hyperemesis gravidarum	Miscarriage, vanishing twin
Anaemia	**Congenital anomalies** (2–3× in MZ twins)

Maternal	Fetal / neonatal
Pre-eclampsia/eclampsia (earlier, more severe), GDM	**Preterm birth** (about 50% <37 weeks) and its complications
Preterm labour, PPROM, polyhydramnios	**FGR**, discordant growth, sFGR
APH: placenta praevia, abruption (after twin 1)	**TTTS, TAPS, TRAP** (MC only)
Malpresentation, cord prolapse	Cord entanglement (MCMA), conjoined twins
PPH (atony), retained placenta	Single fetal death and co-twin injury (MC)
VTE, pressure symptoms	**Intrapartum hypoxia of twin 2**, locked twins
Postnatal depression, financial strain	Higher perinatal mortality (several times that of singletons), cerebral palsy

Treatment- and delivery-related

Complication	Cause	Prevention / action
Trapped twin 2	Uterotonic given after twin 1	No oxytocic until the last baby
Operative delivery, caesarean morbidity	Malpresentation of twin 2, locked twins	CS when twin 1 is non-cephalic; experienced operator
Cord prolapse	ARM before the presenting part is in the pelvis	ARM with a contraction once engaged
Twin 2 hypoxia	Long inter-twin interval	Monitor FHR; aim ≤30 min if not continuously monitored
Prematurity from planned early delivery	MC/MCMA timing	Steroids before planned early birth
Preterm labour, membrane rupture, infection	Amnioreduction, laser	Specialist centres

11. Prognosis

- Perinatal mortality is **several times higher than for singletons**; about **half** of twins are born before 37 weeks.
- **DCDA** twins have the lowest risk; **MC** twins carry most of the fetal risk.
- **TTTS**: without treatment, severe mid-trimester TTTS has a perinatal mortality of **80–90%**; with laser, at least one survivor in about **85–90%**.
- **Single fetal death**: MC survivor death ~15% and neurological injury ~25%; DC ~3% and ~2%.
- **Locked twins**: very high perinatal mortality for twin 1.
- **Recurrence**: DZ twinning tends to recur.

12. Prevention

- **Of twins themselves**: limit the number of embryos transferred in IVF and monitor ovulation induction.
- **Of complications**: early chorionicity scan; enhanced ANC; iron, folate and calcium; **aspirin** where indicated; IPTp-SP and ITN; treat UTIs and malaria; BP and urine at every visit; scan surveillance by chorionicity.
- **Of preterm birth**: routine cerclage, routine progesterone and bed rest do **not** help in twins; treat triggers; steroids when preterm birth is likely.

- **Of intrapartum harm**: deliver at a facility with theatre and blood; CS when twin 1 is non-cephalic (prevents locked twins); two neonatal teams; no oxytocic until the last baby.
- **Of PPH**: oxytocin after the last baby plus a prophylactic infusion; correct anaemia antenatally.
- **Of a future unplanned pregnancy**: family planning before discharge; interpregnancy interval ≥24 months.

13. Special Populations

- **Monochorionic twins** (or chorionicity uncertain): 2-weekly scans from 16 weeks, delivery at 36+0 weeks, fetal medicine referral for TTTS, TAPS, sFGR or co-twin death.
- **MCMA twins**: CS at 32+0–33+6 weeks after steroids.
- **Triplets and higher-order multiples**: delivery at 35+0 weeks, usually by CS; refer to a referral hospital.
- **Previous caesarean section**: one previous CS is not an absolute indication for CS, but most Ugandan units favour repeat CS; two or more previous CS or a classical scar → CS.
- **Women living far from hospital**: admit near term (e.g. 34–36 weeks) for birth preparedness.
- **HIV-positive women**: eMTCT plan and infant prophylaxis for **both** babies.
- **Additional pre-eclampsia risk factor** (first pregnancy, age ≥40, BMI ≥35): aspirin 75–150 mg from 12 weeks.
- **Undiagnosed twins in labour at HC II/III**: deliver twin 1 if imminent, no oxytocic, refer if twin 2 is transverse or delayed.

14. Key Points

High-yield points

- **Chorionicity, not zygosity**, decides risk; read it at 11–14 weeks; if uncertain, manage as MC.
- "Large for dates" in Uganda is twins until proven otherwise: scan.
- Scans: DC 4-weekly from 20 weeks; MC 2-weekly from 16 weeks.
- Discordance: divide by the **larger** twin; ≥25% is significant.
- TTTS: donor DVP <2 cm, recipient >8 cm; Quintero staging; laser 16–26 weeks.
- Extra iron and folate; aspirin if one more risk factor.
- Deliver DCDA 37+0, MCDA 36+0, MCMA 32+0–33+6 by CS, triplets 35+0.
- Vaginal birth if twin 1 is cephalic; CS if twin 1 is non-cephalic.
- **No oxytocic after twin 1**; stabilise, listen, examine, ARM when engaged.
- Uterotonic after the **last** baby, plus an infusion; watch for PPH.

Numbers & doses

Item	Value
MZ twinning rate	About **3.5–4 per 1,000** (constant worldwide)
DZ twinning in sub-Saharan Africa	Often around **18–28 per 1,000** births
MZ split timing	**0–3 days** DCDA; **4–8** MCDA; **8–12** MCMA; **≥13** conjoined
Proportion of MZ twins	DCDA ~25–30%; MCDA ~70–75%; MCMA ~1–2%
Chorionicity scan	**11–14 weeks**; membrane thick >2 mm (DC) vs thin <2 mm (MC)
Two fetal hearts	Heard simultaneously, differing by **≥10 bpm**
Scan schedule	DC **4-weekly from 20 weeks**; MC **2-weekly from 16 weeks**

Item	Value
Discordance	**(larger – smaller) ÷ larger × 100**; significant **≥25%** (some ≥20%)
sFGR	One twin EFW **<10th centile**
TTTS liquor	Donor DVP **<2 cm**; recipient **>8 cm** (some centres >10 cm after 20 weeks)
TTTS incidence	**10–15%** of MCDA twins
TTTS laser window	**16–26 weeks**
TAPS (classic)	MCA-PSV **>1.5 MoM** one twin, **<1.0 MoM** the other
Co-twin death, MC survivor	Death ~**15%**, neurological injury ~**25%** (DC ~3% and ~2%)
Hb checks	Booking, **20–24 weeks**, **28 weeks**
Iron	Ferrous sulphate 200 mg (60 mg elemental) + folic acid 400 µg daily; treatment **120 mg elemental/day**
Aspirin	**75–150 mg nocte from 12 weeks** to 36 weeks (NICE: until birth)
Planned birth (NICE NG137)	DCDA **37+0**; MCDA **36+0**; MCMA **32+0–33+6 (CS)**; triplets **35+0**
Preterm twins	About **half** born before 37 weeks
Steroids	Dexamethasone **6 mg IM 12-hourly × 4** if birth likely at 24 (28)–34 weeks
Blood for twin delivery	**Group and cross-match 2 units**, 16–18 G cannula
Twin 2: wait before oxytocin	About **10 min** without effective contractions → oxytocin infusion (e.g. **2.5–5 IU in 500 mL**, titrated; check unit protocol)
Inter-twin interval	Traditionally aim **≤30 min** if twin 2 is not continuously monitored
Third stage	**Oxytocin 10 IU IM/IV after the last baby** + infusion **20–40 IU in 1 L**
Locked twins	About **1 in 1,000** twin deliveries

Exam traps

- **"Same-sex twins are identical."** Wrong: same-sex DCDA twins may be DZ or MZ; only **different-sex** proves DZ, and only **MC** proves MZ.
- **"DCDA means non-identical."** Wrong: about a quarter of MZ twins are DCDA (split in days 0–3).
- **"Give oxytocin 10 IU IM after twin 1 as part of AMTSL."** Wrong: this can trap twin 2. **No uterotonic until the last baby is born.**
- **"Discordance = difference ÷ smaller twin."** Wrong: divide by the **larger** twin's EFW.
- **"Twins = elective caesarean."** Wrong: if twin 1 is **cephalic**, planned vaginal birth is reasonable (Twin Birth Study); CS for non-cephalic twin 1, MCMA and triplets.
- **"Chorionicity can be read on a 30-week scan."** Unreliable: the lambda sign fades; if in doubt **manage as MC**.
- **"Deliver immediately after an MC co-twin death to protect the survivor."** Wrong: the damage happens at the moment of death; immediate delivery does not prevent it and adds prematurity.
- **"Cerclage or progesterone prevents preterm birth in twins."** Wrong: routine cerclage, progesterone and bed rest do **not** help.

Questions seniors ask

On the ward round

1. **How do you diagnose twins clinically, and how do you confirm them?** Large for dates, multiple poles and parts, a small head for the uterine size, two FHRs differing by ≥10 bpm; **confirm by ultrasound**, with

chorionicity at 11–14 weeks.

2. **What is the chorionicity, and why does it matter?** DC vs MC decides scan frequency, delivery date and specific risks (TTTS, TAPS, sFGR and co-twin death in MC; cord entanglement in MCMA).
3. **Calculate the discordance for 2,400 g and 1,800 g.** (2,400 – 1,800) ÷ 2,400 = **25%**: significant; check for sFGR and Doppler.
4. **What do you prepare before a twin delivery?** Experienced obstetrician, theatre ready, anaesthetist aware, IV access, 2 units cross-matched, both FHRs monitored, **two neonatal resuscitation sets**, labelled cord clamps, oxytocin infusion ready but not running, ultrasound to hand.
5. **Twin 1 delivered; twin 2 is transverse with intact membranes and a normal FHR. What next?** Stabilise, **ECV** to a longitudinal lie; if it fails, an experienced obstetrician performs **internal podalic version and breech extraction**; otherwise **caesarean section**.
6. **Why are twins at risk of PPH, and how do you prevent it?** Overdistended uterus, large placental bed, anaemia, long labour, operative delivery. Oxytocin 10 IU after the last baby, **prophylactic oxytocin infusion**, close monitoring for 1–2 h, TXA if bleeding.
7. **MCDA twins at 20 weeks with DVP 1.5 cm and 9 cm: what is it? TTTS** (Quintero I if the donor bladder is visible): urgent fetal medicine referral, close scans, laser (16–26 weeks) or amnioreduction where laser is unavailable.
8. **When should uncomplicated twins be delivered?** DCDA 37+0, MCDA 36+0, MCMA 32+0–33+6 by CS, triplets 35+0 (NICE NG137).

Viva questions

1. **How do you determine chorionicity?** Ultrasound at 11–14 weeks: lambda sign = DC; T sign = MC; also separate placentas, different sexes (DC), membrane thickness.
2. **What decides whether MZ twins are DC, MC or conjoined?** The day of splitting: 0–3 days DCDA, 4–8 days MCDA, 8–12 days MCMA, >13 days conjoined.
3. **Explain TTTS.** In MC placentas, unbalanced arteriovenous anastomoses shunt blood from donor to recipient: the donor becomes hypovolaemic, oliguric, oligohydramniotic and growth restricted; the recipient becomes hypervolaemic, polyuric, polyhydramniotic, in heart failure and hydropic. Treated by fetoscopic laser at 16–26 weeks.
4. **What do you do immediately after twin 1 is delivered?** No oxytocic; clamp and label the cord; palpate for the lie of twin 2 and stabilise it; check the FHR; VE for presentation and cord; ARM when the presenting part is in the pelvis; oxytocin infusion if contractions are inadequate; manage malpresentation (ECV, internal podalic version + breech extraction, or CS).
5. **When should twins be delivered?** DCDA 37 weeks; MCDA 36 weeks; MCMA 32–34 weeks by CS; triplets 35 weeks.
6. **What are locked twins, and how are they prevented?** Twin 1 breech and twin 2 cephalic with interlocking chins; prevented by CS when twin 1 is non-cephalic.
7. **Why is the risk of PPH increased?** Overdistended uterus contracts poorly, large placental bed, prolonged labour, operative delivery, anaemia.
8. **Why is the co-twin of an MC twin that dies at risk?** Acute exsanguination through placental anastomoses into the low-pressure circulation of the dead twin → death or brain injury.

Mnemonics

> EXAM FAVOURITE
> **"Lambda = Lots (two chorions); T = Together (one chorion)."**

- **Split timing: "3–8–12–13"**: split by day **3** = DCDA; by day **8** = MCDA; by day **12** = MCMA; after day **13** = conjoined.

- **Delivery dates: "37-36-32-35"**: DCDA **37**, MCDA **36**, MCMA **32–33+6** (by CS), triplets **35**.
- **Maternal complications of twins: "HAPPI PPH"**: **H**yperemesis, **A**naemia, **P**re-eclampsia, **P**reterm labour, **I**ncreased malpresentation/operative delivery, **P**olyhydramnios, **P**raevia/abruption, **H**aemorrhage postpartum.
- **TTTS donor vs recipient: "Donor Dries, Recipient Receives (too much)"**: the donor is dry (oligohydramnios, small bladder, FGR); the recipient is flooded (polyhydramnios, big bladder, heart failure).
- **Twin 2 steps: "No Oxy, Palpate, Stabilise, Listen, Examine, Rupture, Push"**.

OSCE tips

- **Abdominal examination station** (large for dates): inspect, measure the SFH, palpate the poles systematically, note "multiple fetal parts", auscultate two fetal hearts, and state that you would confirm with an ultrasound and determine chorionicity.
- **Counselling station (newly diagnosed MCDA twins)**: congratulate, explain chorionicity in simple words ("they share one placenta"), extra scans every 2 weeks and why (TTTS), symptoms to report (rapid abdominal growth, breathlessness), nutrition/iron, delivery plan at a hospital around 36 weeks.
- **Delivery of twin 2** (mannequin/viva): say "**no oxytocin**", stabilise the lie, check the FHR, VE, ARM with a contraction once the presenting part is engaged, start oxytocin infusion if contractions stop, and describe the manoeuvres for a transverse lie.

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Zygosity vs chorionicity: what is the difference?	**Zygosity** = number of fertilised eggs (MZ vs DZ; genetics). **Chorionicity** = number of placentas (DC vs MC). **Chorionicity determines clinical risk.**
2	Day of splitting of a monozygotic embryo and the resulting twin type?	**0–3 days → DCDA** (~25–30%); **4–8 days → MCDA** (~70–75%); **8–12 days → MCMA** (~1–2%); **>13 days → conjoined**.
3	Are all DCDA twins non-identical?	**No.** All DZ twins are DCDA, but ~25–30% of MZ twins are also DCDA (early split). **All MC twins are MZ.** Different sexes = DZ.
4	Lambda (twin-peak) sign vs T sign?	**Lambda sign**: triangular chorionic tissue at the base of the inter-twin membrane = **dichorionic**. **T sign**: thin membrane meeting the placenta at 90° = **monochorionic diamniotic**.
5	Best gestation to determine chorionicity?	**11–14 weeks** (the lambda sign fades later). If uncertain, **manage as monochorionic**.
6	Risk factors for dizygotic twins?	**Black African ethnicity**, maternal family history, **older maternal age** (35–39), **high parity**, tall/heavy women, **ovulation induction/IVF**. (MZ rate is constant ~3.5–4/1,000.)
7	Pathophysiology of TTTS?	MC placenta with **unbalanced arteriovenous anastomoses** → net transfusion donor → recipient. **Donor**: hypovolaemia, oliguria, oligohydramnios ('stuck twin'), FGR. **Recipient**: hypervolaemia, polyuria, polyhydramnios, cardiac failure, hydrops.
8	How common is TTTS?	About **10–15% of MCDA** twin pregnancies. It does **not** occur in DC twins.
9	Quintero staging of TTTS?	**I**: oligo (<2 cm)/poly (>8 cm) sequence, donor bladder visible. **II**: donor bladder not visible. **III**: critically abnormal Doppler. **IV**: hydrops. **V**: death of one or both twins.
10	Treatment of choice for TTTS?	**Fetoscopic laser ablation** of placental anastomoses at **16–26 weeks** (stage II+). Alternatives: serial **amnioreduction**, selective reduction. Laser is not routinely available in Uganda.
11	How is growth discordance calculated, and what is significant?	(Larger EFW − smaller EFW) ÷ larger EFW × 100. **≥20–25%** is significant (NICE: ≥25% refer).
12	Ultrasound surveillance schedule for DC vs MC twins?	**DC**: growth scans every **4 weeks from 20 weeks**. **MC**: every **2 weeks from 16 weeks** (liquor, bladders, EFW, Doppler) to detect TTTS/sFGR/TAPS.
13	Planned timing of delivery (NICE) for uncomplicated DCDA, MCDA, MCMA and triplets?	**DCDA 37+0**; **MCDA 36+0**; **MCMA 32+0–33+6 by CS**; **triplets 35+0** weeks.
14	Mode of delivery for twins based on twin 1's presentation?	**Twin 1 cephalic** → vaginal birth is reasonable (Twin Birth Study). **Twin 1 non-cephalic**, MCMA, conjoined, triplets → **caesarean section**.
15	What must you NOT do after delivery of twin 1?	**Do not give any uterotonic** (oxytocin, ergometrine, misoprostol) until the **last** baby is delivered: it could trap twin 2 (tetanic uterus) or cause placental separation.

#	Question	Answer
16	Steps after twin 1 is delivered?	Clamp and label cord → **palpate lie of twin 2** and stabilise it longitudinal → **check FHR** → VE (membranes, presenting part, cord) → **ARM with a contraction** when the presenting part is in the pelvis → oxytocin infusion if contractions weak → deliver; manage malpresentation (ECV, internal podalic version + breech extraction, or CS).
17	Management of a transverse second twin?	**ECV** (to cephalic or breech) with membranes intact if FHR normal; if unsuccessful, **internal podalic version + breech extraction** by an experienced obstetrician; otherwise **caesarean section**.
18	What are locked twins?	Twin 1 **breech** + twin 2 **cephalic**: chins interlock above the pelvic brim. ~1 in 1,000 twin deliveries. **Prevented by CS when twin 1 is non-cephalic.**
19	Risks to the surviving twin after single fetal death: MC vs DC?	**MC**: co-twin death ~15%, neurological injury ~25% (acute exsanguination through anastomoses). **DC**: ~3% death, ~2% neuro injury; main risk is preterm labour.
20	Maternal complications of multiple pregnancy?	Hyperemesis, **anaemia**, **pre-eclampsia**, GDM, **preterm labour/PPROM**, polyhydramnios, APH (praevia, abruption), **malpresentation**, operative delivery, **PPH**, VTE, postnatal depression.
21	What is TAPS and how is it diagnosed?	**Twin anaemia-polycythaemia sequence**: tiny anastomoses in MC twins → one anaemic, one polycythaemic, **without** liquor discordance. Diagnosed by **MCA peak systolic velocity** Doppler.
22	EFW twin 1 = 2,400 g, twin 2 = 1,800 g. Calculate the discordance. Is it significant?	**(2,400 – 1,800) ÷ 2,400 × 100 = 25% → significant** (≥25%; some units act at ≥20%). Check whether the small twin is <10th centile (sFGR), its umbilical artery Doppler and the chorionicity. Divide by the **larger** twin (600 ÷ 1,800 = 33% is the classic error).
23	A woman has twins on a 28-week scan, same sex, one placental mass, and chorionicity was never assessed. How do you manage her?	**Manage as monochorionic** (the higher-risk assumption): 2-weekly scans (DVP, bladders, EFW, Doppler) at a referral centre, watch for TTTS/sFGR/TAPS, and plan delivery at **36+0 weeks**. Only different sexes or two separate placentas reliably prove DC this late.
24	Checklist before a planned vaginal twin delivery?	Scan for **presentation of both twins**; experienced obstetrician; **theatre** and anaesthetist ready; **16–18 G IV access**, Hb, **group and cross-match 2 units**; monitor **both FHRs**; **two neonatal resuscitation sets** and labelled cord clamps; oxytocin infusion **prepared but not running**; ultrasound to hand for twin 2.
25	How do you confirm on the ward that you are hearing two different fetal hearts?	Two examiners listen **simultaneously** at different points of the abdomen; rates differing by **≥10 bpm** suggest two hearts. Check the **maternal pulse** too. It is unreliable alone: **confirm by ultrasound**.
26	TTTS liquor criteria and where the TTTS work-up is done in Uganda?	MC twins with donor **DVP <2 cm** and recipient **DVP >8 cm** (some centres >10 cm after 20 weeks). Quintero stage by donor bladder, Dopplers, hydrops, death. Specialist scans at **regional/national referral** or private centres; laser (16–26 weeks) is **not routinely available in Uganda**, so amnioreduction, steroids and timely delivery are the usual options.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 28-year-old G3P2 at 12 weeks has an ultrasound showing two gestational sacs with a thin dividing membrane meeting the single placental mass at a right angle, with no tissue between the layers. What is the MOST likely chorionicity?

A Dichorionic diamniotic
B Dichorionic monoamniotic
C Monochorionic monoamniotic
D Monochorionic diamniotic
E Conjoined twins

2. A 30-year-old at 20 weeks with MCDA twins complains of rapid abdominal enlargement and breathlessness over 2 weeks. Ultrasound: twin A deepest vertical pocket 10 cm with a large bladder; twin B deepest pocket 1 cm, bladder not visible, 'stuck' to the uterine wall. Dopplers are normal. What is the diagnosis and Quintero stage?

A Selective FGR, type I
B TTTS stage II
C TTTS stage I
D TAPS
E TTTS stage IV

3. A 26-year-old para 2 is in labour with twins at a HC IV. Twin 1, cephalic, is delivered normally. The midwife reaches for oxytocin 10 IU IM for 'active management of the third stage'. What is the MOST appropriate action?

A Give oxytocin 10 IU IM now
B Give ergometrine 0.5 mg instead
C Proceed to controlled cord traction
D Give misoprostol 600 mcg orally
E Withhold uterotonics, palpate to determine the lie of twin 2 and check its fetal heart

4. After delivery of twin 1 (cephalic) at a regional referral hospital, twin 2 is found to be in a transverse lie with intact membranes and a normal FHR of 140. The obstetrician is experienced. What is the MOST appropriate next step?

A External cephalic version, and if unsuccessful, internal podalic version with breech extraction
B Rupture the membranes immediately
C Start an oxytocin infusion to encourage the fetus to correct its lie
D Wait 2 hours for spontaneous correction
E Give ergometrine to deliver the placenta of twin 1

5. A 32-year-old at 36 weeks with DCDA twins attends the antenatal clinic at Mengo. Twin 1 is breech and twin 2 is cephalic on ultrasound. Both are well grown. What is the MOST appropriate mode of delivery?

A Vaginal delivery with breech extraction
B Induction of labour at 37 weeks
C Elective caesarean section
D External cephalic version of twin 1 at 36 weeks, then induction
E Await spontaneous labour and decide in labour

6. A 24-year-old with uncomplicated DCDA twins, twin 1 cephalic and twin 2 breech, asks when her babies should be born. She is now 34 weeks. What is the BEST advice?

- A Planned birth at 34 weeks
- B Planned birth at 37+0 weeks
- C Planned birth at 36+0 weeks
- D Wait for spontaneous labour up to 41 weeks
- E Planned birth at 32 weeks by caesarean

7. A 29-year-old at 12 weeks has monochorionic monoamniotic twins confirmed on scan. What is the MOST appropriate delivery plan if the pregnancy remains uncomplicated?

- A Vaginal delivery at 37 weeks
- B Caesarean section at 39 weeks
- C Induction at 36 weeks
- D Caesarean section at 32+0 to 33+6 weeks after antenatal corticosteroids
- E Vaginal delivery if twin 1 is cephalic, at 34 weeks

8. A 34-year-old para 5 delivered twin 1 at home 6 hours ago and walked to a HC III. The uterus is at 34 cm, a second fetus is palpable in a longitudinal lie with cephalic presentation, FHR 136, cervix 6 cm, membranes intact, and there are no contractions. There is no theatre at the HC III. What is the MOST appropriate action? Advanced

- A Start an oxytocin infusion and, once the head is in the pelvis, rupture membranes and deliver, with referral if delay or distress
- B Refer to the HC IV without doing anything
- C Give ergometrine to deliver the placenta
- D Perform internal podalic version
- E Give misoprostol 800 mcg sublingually

9. A 31-year-old with MCDA twins has a scan at 28 weeks showing the death of twin B. Twin A is alive with normal liquor. Which statement is MOST accurate? Advanced

- A Immediate delivery prevents neurological damage in twin A
- B Twin A is at no risk because the twins had separate sacs
- C The risks to twin A are the same as in DC twins
- D Maternal DIC is the main risk and requires immediate delivery
- E Twin A has about a 15% risk of death and 25% risk of neurological injury

10. A 27-year-old at 22 weeks with twins has a scan. The two sacs are separated by a thick membrane with a triangular projection of placental tissue at its base. One twin is male and the other female. Which statement is CORRECT?

- A These are monozygotic dichorionic twins
- B These are monochorionic diamniotic twins
- C These are dizygotic, dichorionic diamniotic twins
- D These twins are at risk of TTTS
- E Chorionicity cannot be determined

11. Immediately after vaginal delivery of both twins (combined weight 5.4 kg) at a HC IV, a 30-year-old para 4 bleeds heavily; the uterus is soft and above the umbilicus. She has had oxytocin 10 IU IM. What is the MOST likely cause, and what is the NEXT step?

A Uterine atony: uterine massage, oxytocin infusion, TXA 1 g IV and IV fluids
B Genital tract trauma: suture tears
C Retained placenta: manual removal
D Coagulopathy: give fresh frozen plasma
E Uterine inversion: manual replacement

12. During a twin vaginal delivery, twin 1 (breech) delivers to the umbilicus, then the trunk will not descend further. Abdominal palpation reveals the head of twin 2 in the pelvis. What complication has occurred?

A Cord prolapse
B Conjoined twins
C Shoulder dystocia
D Locked twins
E Uterine rupture

13. A 25-year-old primigravida with DCDA twins books at 11 weeks at a HC IV. BMI 26, BP 110/70, no other medical history. Which is the MOST appropriate prevention plan?

A Iron and folic acid only
B Aspirin 75–150 mg nightly from 12 to 36 weeks, plus iron/folic acid and calcium
C Routine cervical cerclage at 14 weeks
D Vaginal progesterone from 16 weeks
E Bed rest from 28 weeks

14. A 28-year-old with MCDA twins has scans every 2 weeks. At 24 weeks, liquor volumes are normal in both sacs, but the MCA peak systolic velocity is raised in twin A (anaemia) and reduced in twin B (polycythaemia). What is the diagnosis? Advanced

A TTTS stage I
B Rhesus isoimmunisation
C Selective FGR
D TRAP sequence
E Twin anaemia-polycythaemia sequence (TAPS)

15. A 33-year-old at 26 weeks with DCDA twins has a haemoglobin of 8.6 g/dL. She is on one tablet of ferrous sulphate/folic acid daily. Malaria test is negative. What is the MOST appropriate management?

A Continue the current dose and recheck at delivery
B Blood transfusion immediately
C Increase to elemental iron 120 mg daily (e.g. ferrous sulphate 200 mg twice daily) with folic acid, deworm, and recheck Hb in 4 weeks
D Stop iron because it causes constipation
E Start vitamin C only

Answers & explanations

> **1. D** · Monochorionic diamniotic
>
> The **T sign** (thin membrane meeting the placenta at 90°) indicates **MCDA**. **DCDA** shows the **lambda (twin-peak) sign**. **MCMA** has **no dividing membrane**. **Dichorionic monoamniotic** does not exist (the chorion forms before the amnion). **Conjoined twins** share body parts with no membrane.

2. B · TTTS stage II

Oligohydramnios in the donor (<2 cm) and polyhydramnios in the recipient (>8 cm) = **TTTS**. The **donor bladder is not visible** with normal Dopplers = **stage II**. **Stage I** has a visible donor bladder. **Stage IV** requires hydrops. **sFGR** has no oligo/poly sequence. **TAPS** has Hb discordance without liquor discordance.

3. E · Withhold uterotonics, palpate to determine the lie of twin 2 and check its fetal heart

After twin 1, **no uterotonic** should be given until the **last baby** is delivered: it may cause a tetanic contraction trapping twin 2, or placental separation and hypoxia. The next step is to **palpate the lie** of twin 2, stabilise it and **listen to the FHR**. **Ergometrine** and **misoprostol** carry the same danger. **Controlled cord traction** is for the third stage, which has not begun.

4. A · External cephalic version, and if unsuccessful, internal podalic version with breech extraction

For a transverse second twin with intact membranes and a normal FHR, attempt **ECV**; if it fails, an experienced operator performs **internal podalic version + breech extraction**; CS is the alternative. **ARM in a transverse lie** risks cord or arm prolapse. **Oxytocin** with a transverse lie risks obstruction and uterine rupture. **Waiting 2 hours** increases hypoxia of twin 2. **Ergometrine** is contraindicated before twin 2 is delivered.

5. C · Elective caesarean section

When **twin 1 is non-cephalic**, **caesarean section** is recommended; breech/cephalic carries the risk of **locked twins**. **Vaginal breech delivery** and **induction** do not remove this risk. **ECV** is not performed in twin pregnancies. **Waiting for labour** risks an emergency situation and locked twins.

6. B · Planned birth at 37+0 weeks

Uncomplicated **DCDA** twins: planned birth at **37+0 weeks** (NICE); continuing increases stillbirth risk. **36+0 weeks** is for **MCDA**. **32–33+6 weeks by CS** is for **MCMA**. **34 weeks** is premature without indication. **Waiting to 41 weeks** increases stillbirth risk.

7. D · Caesarean section at 32+0 to 33+6 weeks after antenatal corticosteroids

MCMA twins share one sac with near-universal **cord entanglement** and a risk of sudden death, so **planned CS at 32+0–33+6 weeks** (after steroids) is recommended. **Vaginal delivery** risks cord compression of twin 2 and entanglement. Waiting to **36–39 weeks** increases the risk of sudden intrauterine death.

8. A · Start an oxytocin infusion and, once the head is in the pelvis, rupture membranes and deliver, with referral if delay or distress

Twin 2 is **longitudinal and cephalic** with a normal FHR and an open cervix, so **augmenting with an oxytocin infusion** (dilute and titrated) to restore contractions, then ARM once the head is in the pelvis, is appropriate; refer urgently if there is delay, fetal distress or a malpresentation. **Ergometrine** and **high-dose misoprostol** before delivery of twin 2 risk a tetanic uterus and rupture. **Referral without action** is reasonable only if the birth cannot be safely conducted; here a cephalic baby can be delivered. **Internal podalic version** is for a transverse lie.

9. E · Twin A has about a 15% risk of death and 25% risk of neurological injury

In **MC twins**, the survivor has about a **15% risk of death** and **~25% risk of neurological injury** from acute exsanguination into the dead twin through anastomoses. **Immediate delivery** does not prevent this, because the injury happens at the time of death, and it adds prematurity. **DC twins** have much lower risks (about 3% and 2%). **Maternal DIC** is rare and delayed. Separate **sacs** do not matter; the shared **placenta** does.

10. C · These are dizygotic, dichorionic diamniotic twins

The **lambda sign** indicates **dichorionic** twins, and **different sexes** prove **dizygotic** twins. MZ twins are always the **same sex**. They are not MC (no **T sign**). **TTTS** occurs only in **MC** twins. Chorionicity **can** be determined here.

11. A · Uterine atony: uterine massage, oxytocin infusion, TXA 1 g IV and IV fluids

A **soft, large uterus** after an **overdistended twin pregnancy** = **atonic PPH**. Next: **massage**, **oxytocin infusion** (20–40 IU in 1 L), **TXA 1 g IV**, IV fluids, catheterise and escalate (E-MOTIVE bundle). **Trauma** gives bleeding with a contracted uterus. **Retained placenta** is not suggested (placentas delivered). **Coagulopathy** is less likely immediately. **Inversion** presents with shock and an absent fundus.

12. D · Locked twins

Locked twins: twin 1 **breech** and twin 2 **cephalic**, with interlocking chins preventing delivery of twin 1's head. This is why CS is advised when twin 1 is non-cephalic. **Cord prolapse** does not arrest the trunk. **Shoulder dystocia** follows delivery of the head in a cephalic birth. **Conjoined twins** are usually diagnosed antenatally and delivered by CS. **Uterine rupture** causes pain, shock and loss of contractions.

13. B · Aspirin 75–150 mg nightly from 12 to 36 weeks, plus iron/folic acid and calcium

Multiple pregnancy and first pregnancy are **two moderate risk factors** for pre-eclampsia, so **aspirin 75–150 mg from 12 to 36 weeks** is indicated, plus enhanced iron/folate and calcium. **Iron and folate alone** miss pre-eclampsia prevention. **Routine cerclage**, **progesterone** and **bed rest** do not prevent preterm birth in twins.

14. E · Twin anaemia-polycythaemia sequence (TAPS)

TAPS: small anastomoses cause a slow transfusion, giving **Hb discordance without liquor discordance**; diagnosed by MCA-PSV. **TTTS** requires the oligo/polyhydramnios sequence. **sFGR** concerns weight discordance. **TRAP** involves an acardiac twin. **Rh isoimmunisation** would affect both twins similarly (and requires an Rh-negative mother with antibodies).

15. C · Increase to elemental iron 120 mg daily (e.g. ferrous sulphate 200 mg twice daily) with folic acid, deworm, and recheck Hb in 4 weeks

Twins increase iron and folate demand. With **moderate anaemia (Hb 7–9.9)** and no urgent symptoms, give **treatment-dose iron (120 mg elemental daily)** with folic acid, deworm (mebendazole after the first trimester), and **recheck** (expect about 1 g/dL rise every 2–3 weeks); consider IV iron if no response. **Continuing the prophylactic dose** is inadequate. **Transfusion** is for severe anaemia (Hb <7) with decompensation or near delivery. **Stopping iron** worsens anaemia. **Vitamin C alone** is insufficient.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Large for dates at 24 weeks

A 34-year-old para 5 attends her first ANC visit at a HC III in Mpigi at 24 weeks by LNMP. She has had excessive vomiting earlier in pregnancy and now feels breathless on exertion. Symphysis-fundal height is 31 cm. Conjunctivae are pale. BP 118/74.

1. What is your differential diagnosis for a fundal height larger than dates?

> **Multiple pregnancy** (top in a para 5 aged 34 with hyperemesis), wrong dates, polyhydramnios, macrosomia/GDM, fibroids, molar pregnancy (less likely at 24 weeks with fetal parts).

2. What history and examination would support twins?

> History: maternal **family history of twins**, fertility drugs, hyperemesis, rapid abdominal growth, 'feeling movements everywhere'. Exam: many fetal parts, **two fetal poles** (two heads), a small head relative to the uterus, **two fetal hearts** heard simultaneously by two examiners differing by ≥10 bpm.

3. What investigations and management do you arrange at the HC III?

> **Hb** (pallor, breathlessness), urinalysis, mRDT, HIV/syphilis per ANC. **Refer for an ultrasound** (number, chorionicity, anomalies, growth, liquor, placenta) and to a **hospital ANC clinic**. Start iron/folate (treatment dose if Hb is low), IPTp-SP, Td, ITN, deworming. Health education on danger signs; birth preparedness at a facility with theatre.

4. Ultrasound at the hospital: live twins, separate posterior and fundal placentas, thick membrane, one male and one female. Hb 9.2 g/dL. What does this mean and what is your ANC plan?

> **DCDA, dizygotic twins** (different sexes). Plan: iron 120 mg elemental daily + folic acid, recheck Hb in 4 weeks; **aspirin 75–150 mg** to 36 weeks (twins + another moderate risk factor, and many give to all twins); calcium; ANC every 2 weeks, then weekly from 32 weeks; BP and urine each visit; **growth scans every 4 weeks**; steroids if preterm birth threatens (28–34 weeks); plan **birth at 37 weeks** at the hospital; consider admission near term as she lives far away.

5. At 35 weeks she presents with a headache and BP 158/104, proteinuria 2+. What do you do?

> Pre-eclampsia (more common and earlier in twins). Check for severe features (headache is one): FBC, LFT, creatinine. Control BP (nifedipine), **$MgSO_4$** if severe features, fluid balance, fetal assessment of both twins. With severe features at 35 weeks: **deliver** after stabilisation; the mode depends on the presentation of twin 1 (vaginal if cephalic). Prepare for PPH.

Take-away: Large for dates in a multiparous Ugandan woman is twins until proven otherwise: confirm by ultrasound, determine chorionicity, treat anaemia and plan delivery in a hospital.

Case 2: Delivering the second twin

A 29-year-old para 2 with known DCDA twins at 37+2 weeks is in active labour at Mengo Hospital. Admission scan: twin 1 cephalic, twin 2 breech. Twin 1 has just been delivered spontaneously and is crying well.

1. What should have been prepared before the birth?

> Experienced obstetrician present, midwife, anaesthetist informed, theatre available; **two neonatal resuscitation sets/teams**; IV access (16–18 G), FBC, **group and cross-match**; both FHRs monitored; oxytocin infusion ready but not running; ultrasound machine in the room; labelled cord clamps.

2. What are your immediate actions after twin 1 is delivered?

> Clamp and cut the cord, **label 'Twin 1'**, hand the baby over. **No uterotonic. Palpate** (± ultrasound) the lie and presentation of twin 2; an assistant **stabilises the lie longitudinally**. **Auscultate twin 2's FHR. VE**: membranes, presenting part, exclude cord prolapse.

3. Twin 2 is now in a transverse lie, membranes intact, FHR 145. What do you do?

> Attempt **ECV** (to breech or cephalic, whichever is easier) with membranes intact while the uterus is relaxed. If it fails, the experienced obstetrician performs **internal podalic version** (grasping a foot through intact membranes) followed by **breech extraction**. If neither is possible, or fetal distress/cord prolapse occurs → **emergency caesarean section**.

4. ECV converts twin 2 to breech, but contractions have stopped and 15 minutes have passed. FHR is 140. What next?

> Start an **oxytocin infusion** (dilute, titrated) to re-establish contractions; when the **breech is in the pelvis**, **ARM** during a contraction; conduct an **assisted breech delivery**. Continue FHR monitoring; if distress, proceed to breech extraction. Aim to deliver without unnecessary delay (historically within 30 minutes).

5. Both babies are delivered. What do you do in the third stage and afterwards?

> **Oxytocin 10 IU IM/IV** after the last baby, controlled cord traction, and a **prophylactic oxytocin infusion** (e.g. 20–40 IU in 1 L over 4 hours). Examine placentas and membranes (confirm DC). **Monitor closely for PPH** (fundus, lochia, pulse, BP every 15 min for the first hour). TXA 1 g IV if bleeding. Check Hb, continue iron. Support breastfeeding of both twins, KMC if low birth weight, and counsel on family planning.

Take-away: After twin 1: no oxytocic; palpate, stabilise, listen, examine; correct a malpresentation; augment if needed; and give the uterotonic only after the last baby.

Case 3: Monochorionic twins with a tense abdomen

A 23-year-old primigravida with MCDA twins (T sign at 12 weeks) presents at 19 weeks to a regional referral hospital with rapidly increasing abdominal size, breathlessness and lower abdominal tightness over 10 days. She missed her 16-week scan because of transport costs.

1. What is your leading diagnosis and why?

> **Twin-twin transfusion syndrome**: MCDA twins share a placenta with vascular anastomoses. Rapid abdominal growth and breathlessness suggest **polyhydramnios** in the recipient. Other causes of polyhydramnios (diabetes, anomalies) are less likely to produce this picture in MC twins.

2. What ultrasound findings will you look for?

> Deepest vertical pocket: **recipient >8 cm**, **donor <2 cm** ('stuck twin'); bladders (donor bladder visible or not); EFWs and discordance; **Dopplers** (umbilical artery end-diastolic flow, ductus venosus a-wave, umbilical vein pulsations); signs of **hydrops**; cervical length (preterm labour risk).

3. The scan shows recipient DVP 11 cm, donor DVP 1 cm, donor bladder not visible, normal Dopplers, no hydrops. Stage it and explain the options.

> **Quintero stage II.** The treatment of choice is **fetoscopic laser ablation** of anastomoses (16–26 weeks), which is not routinely available in Uganda (referral abroad may be possible for some families). Alternatives: **serial amnioreduction** of the recipient sac to relieve symptoms and reduce preterm labour risk; close surveillance; **selective reduction** (rarely available). Untreated severe TTTS has a very high mortality. Counsel honestly and compassionately.

4. She chooses amnioreduction and surveillance. At 27 weeks the recipient develops ascites and pericardial effusion. What does this mean and what is your plan?

> **Hydrops = stage IV.** With viability reached at a referral centre, give **dexamethasone 6 mg IM 12-hourly × 4**, **$MgSO_4$** neuroprotection (<32 weeks) if delivery is planned, discuss with NICU, and consider delivery (usually **caesarean section**), as the risk of intrauterine death of one twin (which endangers the other through the anastomoses) is high. Individualise with the family and senior team.

5. What lesson about her antenatal care does this case teach, and what is the recommended surveillance for MC twins?

> MC twins need scans **every 2 weeks from 16 weeks** (liquor, bladders, growth, Doppler) so that TTTS is detected early when outcomes are better. Barriers such as **transport costs** must be addressed at booking (birth preparedness, social support, health worker follow-up).

Take-away: Monochorionic twins share a placenta: 2-weekly scans from 16 weeks detect TTTS early, and polyhydramnios in one sac with a stuck co-twin is TTTS until proven otherwise.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.